Summary Notes:
Fundamentals of Cloud Computing

1. Introduction

Overview

Here's an overview of what was covered in the session:

1. Overview of Cloud Computing Concepts
    - Definition and basic concepts
    - Key characteristics of cloud computing
    - Historical context and evolution of cloud computing
2. Types of Cloud Services
    - Introduction to service models
    - Infrastructure as a Service (IaaS)
    - Platform as a Service (PaaS)
    - Software as a Service (SaaS)
3. Advantages and Disadvantages of Cloud Computing
    - Benefits of using cloud computing
    - Its potential drawbacks and challenges

2. Overview of Cloud Computing Concepts

2.1 Cloud computing

Cloud computing is the pay-per-use on-demand access to computer resources via the internet (the cloud), including virtual or physical servers, networking, data storage, application development tools and AI-powered analytic tools.

With cloud computing, you rent your IT rather than purchasing it. Instead of making significant upfront investments in databases, software and hardware, businesses choose to access these through the internet or the cloud and pay only for the computing power they need.

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2.2 Key characteristics of cloud computing

- **On-demand self-service**: Cloud computing providers provide consumers access to APIs so they can grow their current resources or request new ones as needed. Terraform and Ansible, two Infrastructure as Code technologies, allow teams to easily automate their infrastructure provisioning.

- **Broad network access**: One of the main benefits of cloud computing is that it provides geographically dispersed physical hardware, enabling businesses to selectively allocate hardware to certain locations, providing the best possible end-user experience.

- **Resource pooling**: An on-demand, dynamic allocation and division of computing resources is seen in cloud infrastructure platforms. Cloud hosts optimise their physical machines for maximum usage, as they are shared among several tenants and dynamically deployed.

- **Rapid elasticity**: The cloud architecture may dynamically scale up and down, allowing customers to request that computing resources be autoscaled in response to traffic demands. Elasticity can be achieved on two levels: one at the machine level, where resource allocation maximises the machine's capabilities, and another at the multi-machine level, where an application adapts to multiple computers on a network automatically.

- **Measured service**: Cloud providers offer comprehensive usage metrics to convey costs incurred from usage. For instance, Amazon Web Services (AWS)provides records of how much each service category has been used, measured either by the hour or by the day. These providers typically adopt a pay-as-you-go billing approach similar to utilities, ensuring customers are billed accurately based on their actual computing resource consumption.

2.3 Historical context and evolution

2.3.1 Early concepts

In the early 2000s, when the term 'the cloud' first appeared, it was understood by only a small group of people with specialised knowledge or interest. The concept of remotely accessing computing resources, rather than relying solely on on-premise IT infrastructure, seemed like something out of science fiction. However, the actual impact was far-reaching, fundamentally transforming technology and business practices.

The roots of cloud computing trace back to the start of the 1960s, when Dr. J.C.R. Licklider, an computer scientist and psychologist often hailed as the ‘father of cloud computing,’ introduced early concepts of

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global networking. He envisioned an Intergalactic Computer Network in his memos, proposing a future where everyone on Earth would be interconnected via computers, accessing information from any location.

2.3.2 Emergence of cloud computing

The popularity of the cloud grew as companies began to understand its benefits and services better. In 1999, Salesforce successfully implemented cloud computing, pioneering the concept of using the internet to deliver software programs to end users. Amazon Web Services (AWS) launched computing services and cloud-based storage in 2002, and it introduced Elastic Compute Cloud (EC2) in 2006, allowing users to rent virtual computers for their applications. During the same year, Google launched Google Apps Suite (now Google Workspace), a suite of SaaS applications for productivity. Microsoft officially introduced Windows Azure in 2010 and entered the SaaS market with Office 365 in 2011.

Despite the challenges faced by small businesses in 2020, cloud operators thrived. Enterprise spending on Infrastructure as a Service (IaaS) reached $37 billion in the fourth quarter and $130 billion for the entire year, a 35% increase from the previous year, according to Synergy Research. In contrast, spending on data centre hardware and software by enterprises fell by 6% to under $90 billion. Gartner also reported robust growth in cloud enterprise spending, which increased by 43% in 2020 and an additional 29% in 2021 to reach $81 billion. It is predicted that around 45% of IT spending would transition from internal infrastructure to cloud services by 2024.

3. Types of Cloud Services

3.1 Introduction to service models

The three main types of cloud services are Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS). These services offer different levels of control and responsibility for customers within their cloud environments. There isn't a universal approach to cloud adoption; the key is to find the right solution that meets your business needs. It's also common for organisations to use a combination of these services.

3.2 Infrastructure as a Service (IaaS)

3.2.1 Definition and examples

Infrastructure-as-a-Service (IaaS) is a fundamental cloud service which lets organisations rent infrastructure of IT such as operating systems, networks, storage and servers from a cloud provider. IaaS allows scaling of resources up or down as needed, minimising the need for large upfront capital expenditures or excess on-premises infrastructure to handle occasional spikes in usage.

Examples of IaaS products would be Microsoft Azure and Amazon Web Services.

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3.2.2 Key features and use cases

IaaS allows reservation of machines that are pre-configured such as email servers, databases and load balancers, for various specialised tasks.

Businesses can implement IaaS for migrating their applications and data to the cloud. For instance, a small enterprise instead of hosting on premises, might move its web servers, emails and files to the cloud.

IaaS can also be used for disaster recovery, as cloud providers store redundant backups across multiple data centres. This ensures that data remains safe even if one data centre experiences issues, allowing businesses for data recovery if it is destroyed by a natural disaster, accidentally deleted or ransomed.

DevOps teams can utilise IaaS as a foundational platform to build a DevOps toolchain, incorporating various third-party tools.

3.3 Platform as a Service (PaaS)

3.3.1 Definition and examples

Platform as a Service (PaaS) is a cloud infrastructure layer built on top of IaaS that supplies the resources needed to create user-level applications and tools. PaaS provides essential infrastructure including servers, database management systems, development tools, storage resources, networking and middleware.

Examples of PaaS would be SAP Cloud or Google App Engine.

3.3.2 Key features and use cases

PaaS provides customers with the developer tools necessary to build and manage mobile and web applications without the need to invest in or maintain the underlying infrastructure.

Nowadays, PaaS is often based on containers, a virtualised compute model that eliminates the need to virtualise hardware and manage separate operating system instances, simplifying the virtual server process. Containers virtualise the operating system, allowing developers to package applications with only the OS services they require, enabling them to run on any platform without modification and without needing middleware.

3.4 Software as a Service (SaaS)

3.4.1 Definition and examples

Software as a Service (SaaS) provides on-demand software applications over the internet, often via subscription. Cloud providers host the applications and handle the software upgrades and security patches.

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Examples of SaaS include Microsoft 365, Dropbox, and DocuSign, as well as productivity tools and webmail applications such as Confluence and Jira, chat applications monitoring tools, analytics tools and more.

3.4.2 Key features and use cases

SaaS is the ideal solution for business as it allows them to quickly adopt the latest technology. Customers can scale services to accommodate changing workloads, adding more features or services as they grow. SaaS offers the following advantages besides cost savings and scalability benefits of the cloud:

- **Automatic upgrades:** As the cloud service provider updates the software, users also gain access to new features automatically sans the need for on-premises upgrades.
- **Protection from data loss:** Users data is not lost if their device crashes or breaks, since SaaS stores application data in the cloud.

SaaS is beneficial for small companies which lack the IT resources or financial resources to deploy the latest technology. They avoid the costs and labour associated with maintaining their own hardware and the high upfront costs of software.

4. Advantages of Cloud Computing

There are several benefits to the unique properties of cloud infrastructures. The key cloud computing benefits are listed.

Cost efficiency

Businesses using cloud resources don't need to purchase hardware assets of their own. In addition to saving on hardware costs, cloud providers work towards maximising and optimising usage of hardware. This turns computing resources and hardware into a commodity, with cloud providers competing to provide the most cost-effective solutions.

Scalability and flexibility

Cloud computing, being inherently elastic, allows organisations to scale resources as needed. Cloud applications can automatically expand or reduce their infrastructure as a result of auto-scaling, as a response to fluctuations in traffic

Increased security

Private cloud provides firewalled, isolated infrastructure which enhances security. Additionally, a range of security technologies and mechanisms to support the development of secure applications is offered by the cloud providers. There are also tools provided to manage user access effectively as user access control is a critical security consideration.

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Accessibility and mobility

Instant and on-demand access to all the applications and files essential for running the business is possible from virtually any location worldwide with the cloud. This ensures a user experience which is consistent across multiple branch locations or offices by empowering employees to maintain productivity regardless of their geographical location.

Disaster recovery and data backup

Developing a disaster recovery plan in cloud computing enables rapid backup of entire servers. Information, applications and systems are consolidated into a single software block or virtual server, facilitating comprehensive backups. This approach ensures backup of critical data and reduces downtime during recovery.

Virtual servers provide exceptional flexibility and resilience, enabling quick relocation to different data centres within minutes. Virtual servers are not dependent on specific hardware unlike traditional servers making them easily recoverable and highly portable. Recovery times can be configured to take just minutes in a disaster scenario, ensuring continuous online operations for businesses.

Increased speed and agility

Cloud computing enables organisations to deploy applications for enterprises within minutes instead of waiting for days or weeks. This capability empowers users, especially development teams and DevOps to efficiently utilise cloud-based software and infrastructure support.

5. Disadvantages of Cloud Computing

Downtime and reliability issues

Cloud services are susceptible to downtime, which can disrupt operations. Even leading cloud service providers may encounter downtime resulting from natural disasters. Connectivity and performance could also be affected from technical issues. During such incidents, access to cloud services may be temporarily unavailable until the issues are resolved. While cloud providers strive for high availability, no service can promise 100% uptime.

Privacy and security concerns

While cloud service providers adhere to industry certifications and stringent security standards, storing critical files and data with external providers carries inherent risks. Security and privacy must be prioritised, especially concerning sensitive information.

Insufficient cloud security measures can lead to data leaks across cloud networks, potentially resulting in malware attacks, breaches of contracts and intellectual property theft. Hackers may also compromise how companies deliver services to their customers or end-users.

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Limited flexibility and control

Cloud users have minimal control, as the cloud service provider owns, monitors and manages the cloud infrastructure. Cloud users may have reduced control over the functionality and execution of services within the cloud-hosted infrastructure, depending on the specific service. Customers maintain control over their applications, services and data, but might not have equivalent control over the underlying infrastructure managed by the cloud provider.

Vendor lock-in

Vendor lock-in is often seen as a drawback of cloud computing. Seamless migration between cloud services is not fully developed, making it challenging for organisations to transfer their services. Variations between vendor platforms may introduce obstacles during migration, potentially leading to increased expenses and configuration complexities.

Cost concerns

Implementing cloud solutions for small-scale or short-term projects can be perceived as costly. While pay-as-you-use cloud services offer flexibility and reduce upfront hardware expenses, the overall cost may exceed initial expectations.

6. Summary and Key Takeaways

Throughout this session, we have explored several crucial aspects of cloud computing. Here are the key takeaways:

1. Overview of Cloud Computing Concepts
    - Definition and Basic Concepts: Cloud computing refers to the provision of on-demand access to a variety of computing resources via the internet. Users can rent IT infrastructure and services rather than purchasing them outright.
    - Key Characteristics: Cloud computing is characterised by several key features including on-demand service, resource pooling, broad network access, measured service and rapid elasticity.
2. Types of Cloud Services
    - Infrastructure as a Service (IaaS): Offers virtualised computing resources over the internet, exemplified by AWS and Microsoft Azure, providing scalable and flexible IT infrastructure.
    - Platform as a Service (PaaS): Provides developers with software and hardware tools over the internet, such as Google App Engine, enabling them to build, deploy and test applications without requiring to handle the underlying infrastructure.
    - Software as a Service (SaaS): Delivers software applications over the internet through a subscription model, such as Microsoft 365 and Google Workspace, enabling users to access software without the burden of managing installations and updates.
3. Advantages of Cloud Computing

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- Cost Efficiency: Reduces the need for large capital expenditures by allowing pay-per-use pricing models.
- Scalability and Flexibility: Enables businesses to scale resources according to demand.
- Increased Security: Offers advanced security features and disaster recovery options.
- Accessibility and Mobility: Provides access to data and applications from anywhere, enhancing productivity.
- Disaster Recovery and Data Backup: Ensures data redundancy and quick recovery times.

4. Disadvantages of Cloud Computing
    - Downtime and Reliability Issues: Potential for service disruptions that can affect business operations.
    - Security and Privacy Concerns: Risks associated with storing sensitive data on external servers.
    - Limited Control and Flexibility: Reduced control over certain aspects of the infrastructure.
    - Vendor Lock-in: Challenges associated with migrating between different cloud providers.
    - Cost Concerns: Potential for higher-than-expected costs, especially for small-scale or short-term projects.

By understanding these core concepts, service models, benefits, and challenges, you are now better equipped to make informed decisions about utilising cloud computing in your personal or professional environment.

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